

Incentive Mechanisms for Online Task Offloading With Privacy-Preserving in UAV-Assisted Mobile Edge Computing

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Abstract—Unmanned aerial vehicles (UAVs) have emerged as a promising technology to provide low-latency mobile edge computing (MEC) services. To fully utilize the potential of UAV-assisted MEC in practice, both technical and economic challenges need to be addressed: how to optimize UAV trajectory for online task offloading and incentivize the participation of UAVs without compromising the privacy of user equipment (UE). In this work, we consider unique features of UAVs, *i.e.*, high mobility as well as limited energy and computing capacity, and propose privacy-preserving auction frameworks, *Ptero*, to schedule offloading tasks on the fly and incentivize UAVs' participation. Specifically, *Ptero* first decomposes the online task offloading problem into a series of one-round problems by scaling the UAV's energy constraint into the objective. To protect UE's privacy, *Ptero* calculates UAV's coverage based on subset-anonymity. At each round, *Ptero* schedules UAVs greedily, computes remuneration for working UAVs, and processes unserved tasks in the cloud to maximize the system's utility (*i.e.*, minimize social cost). Theoretical analysis proves that *Ptero* achieves truthfulness, individual rationality, computational efficiency, privacy-preserving and a nontrivial competitive ratio. Trace-driven evaluations further verify that *Ptero* can reduce the social cost by up to 116% compared with four state-of-the-art algorithms.

Index Terms—UAV trajectory, task offloading, online algorithm, incentive mechanism.

I. INTRODUCTION

WITH the development of 5G, computing-intensive applications running on user equipment (UE), such as online games, VR/AR, [2], requiring low latency and high

computation power [3]. To support such applications, mobile edge computing (MEC) which processes offloading requests from UEs in edge servers at base stations (BSs) has recently emerged as a new computing paradigm [4], [5]. Unfortunately, the infrastructure-based MEC generates high deployment cost, and may not work effectively in rural areas without sufficient infrastructures or urban areas in peak hours/ natural disasters [6]. In recent years, Unmanned Aerial Vehicle (UAV) has received extensive attention due to its high mobility and flexible deployment. UAVs can effectively improve the computing capacity and extend the coverage of MEC [7]. For example, on July 21, 2021, Mihe town in China was submerged due to heavy rainfall. In its communication interruption area, the mobile base station enabled by UAVs provided a mobile signal coverage of about 50 square kilometers for five hours. A total of 3572 users were connected, and the maximum number of users accessing at one time was 648 [8].

To fully realize the potential of UAV-assisted MEC in practice, both *technical* and *economic* challenges need to be addressed. **First**, on the technical side, one UAV cannot provide continuous computing services for a long time because the amount of energy obtained from one charge is limited [7]. Therefore, it is essential to optimize UAV trajectory to serve dynamic offloading requests from UEs under the UAVs' battery capacity constraint. Furthermore, when an emergency occurs, hundreds of UAVs from different organizations need to cooperate to support long-term and continuous services. Collaborative task offloading between UAVs is non-trivial because UAVs from different organizations have heterogeneous computing and energy capacities. For example, In the Henan rescue operation in July 2021, the DJI emergency rescue alliance mobilized 37 drone rescue teams, including 255 drones, to provide aerial sensing and lighting services [9]. **Second**, on the economic side, UAVs process offloading tasks with their own cost, *e.g.*, energy and computing resource consumption [10]. It is difficult to quickly gather enough *voluntary* UAVs in an emergency situation at the same time. Therefore, incentive mechanisms which pay remuneration to working UAVs are necessary for enabling UAV-assisted MEC. **Furthermore**, one UE's private information (*e.g.*, the location, offloading requests) infers the UE owner's hobbies or preferences [11]. This information should be hidden from UAVs to prevent the risk of privacy leakage [4]. The trajectory optimization is influenced by the location information of UEs, and will affect the energy consumption of UAVs. The system's utility is determined by UAVs and UEs. These challenges are

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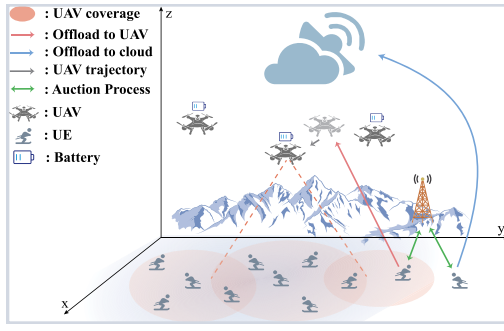


Fig. 1. An illustration of UAV-assist offloading.

coupled with each other. Hence, both technical and economic challenges should be addressed without compromising the privacy of UEs.

There have been some efforts to solve the above two challenges. For the technical side, existing studies on UAV-assisted MEC focus on minimizing the overall UEs'/UAVs' energy consumption [12] or the system delay [13]. For the economic side, there are only a few related studies adopting fixed pricing [6]. However, fixed pricing cannot capture the changing relationship between the supply from UAVs and the demand from UEs. Consequently, overpricing and underpricing routinely occur, jeopardizing the UAV's profit as well as the system's utility. Incentive mechanisms can stimulate everyone in the system to find the most efficient solution, which prompts the system in a long run. Other related papers in incentive mechanism design are either to maximize the individual UAV's profit [10] or the provider's revenue [14] and fail to consider the whole system's utility. Moreover, they incentivize the UAVs to serve or not to serve, and fail to optimize the trajectory of UAVs. We will discuss it in detail in Sec. II. To the best of our knowledge, none of them consider the limited capacity of the UAV's battery in the long term and protect the UE's private information.

To overcome the aforementioned challenges, we capture important features of UAVs, *i.e.*, mobility as well as limited energy and computing capacity, and study the online incentive mechanism design for task offloading with privacy-preserving in UAV-assisted MEC. In particular, we consider an emergency situation, *e.g.*, sports event (as shown in Fig. 1), where a base station (BS) needs to hire UAVs to assist offloading. We utilize auction to incentivize UAVs since it is a natural approach to balance supply and demand and automatically discover the right price so that BS can select the most cost-efficient UAVs. We optimize the system's utility in the long run (*i.e.*, social cost), and decide the trajectories of UAVs on the fly under UAVs' battery capacity constraints. Our main contributions are summarized as follows.

- We formulate the social cost minimization problem of task offloading in UAV-assisted MEC as an integer linear programming problem, which determines the two types of decisions (offloaded to UAVs or the remote cloud) under UAVs' battery capacity constraints. The problem is NP-hard even in offline settings. The challenge further escalates when tackling the online scenario and privacy-preserving.
- We design a privacy-preserving auction framework *Ptero*. *Ptero* first decomposes the online problem into a series of one-round winner determination problems by scaling the UAV's energy constraint into the objective. To protect

the location information of UEs, *Ptero* next calculates each UAV's service scheme based on subset-anonymity. At each round, *Ptero* calls A_{winner} to schedule UAVs based on a greedy approach and process leftover tasks in the cloud to minimize the total cost. A payment strategy is proposed to calculate the remuneration for UAVs based on the critical value.

- We apply the primal-dual theory to prove that *Ptero* achieves a bounded competitive ratio with a small loss in online decomposition. We also prove that the one-round auction A_{winner} is α -approximate. We further prove that *Ptero* guarantees truthfulness, individual rationality, privacy-preserving, and computational efficiency.
- We extend our model to satisfy more general scenarios where the UE requests to offload a large amount of data such that a single UAV can not process the request completely in one slot. We propose a new framework, *Ptero-M*, to solve it. *Ptero-M* modifies the auction metric and selects suitable UEs for UAVs in an iterative fashion. *Ptero-M* also achieves truthfulness, individual rationality, privacy-preserving, and computational efficiency.
- We evaluate the performance of *Ptero* and *Ptero-M* based on real-world traces. Some promising results are observed: 1) A_{winner} achieves a low approximation ratio (≈ 1.2) and can reduce the social cost by up to 12% and 51% compared to *Trac* [15] and *ODSH* [16]; 2) In the online settings, *Ptero* achieves a small competitive ratio (≈ 1.4) and reduces the social cost by up to 116%, 17.8%, and 4.5% compared with two state-of-art scheduling strategies [10], [16] and a naive version of *Ptero*; 3) A_{winner} makes decisions in less than 80ms while *Ptero* executes the online scheduling decisions in less than 160s for 45 rounds; 4) *Ptero-M* reduces the social cost by up to 18% and 14% compares with the other two algorithms, Non-Divided-Offloading, and *ODSH* [16]; 5) Both *Ptero* and *Ptero-M* achieve a low information leakage rate, which is around 2% in the default setting.

In the rest of the paper, we review related work in Sec. II. The auction model for UAV-assisted MEC is introduced in Sec. III. We present our online auction framework, *Ptero*, in Sec. IV and analyze its theoretical performance in Sec. V. Sec. VI illustrates the solution for a general scenario and presents another auction framework *Ptero-M*. Sec. VII shows the trace-driven simulation results and Sec. VIII concludes the paper.

II. RELATED WORK

UAV-assisted MEC. The UAV-assisted task offloading in MEC has been studied under different objectives. Li et al. [13] consider the popularity of caches in the UAVs to optimize their flight strategy, aiming to minimize the system service delay. Zhao et al. [12] focus on the total length of trajectories of UAVs and propose a multi-agent stochastic game to reach the optimal route. Zhang et al. [17] propose a game theory-based scheme to jointly optimize the weighted values of delay and energy. In [5], UAV energy consumption and service delay are iteratively optimized by SCA-based algorithms. Targeting on throughput maximization, Ning et al. [18] develop an average throughput maximization-based auction and a dynamic task admission algorithm. To minimize the service delay under

energy constraints, Liao et al. [19] propose HOTSPOT, which searches the optimal location of UAVs and the offloading balanced scheme. There is also other literature focusing on the cover rate of users [7], the number of deployed UAVs [20], and average throughput [21]. All of these papers focus on technical challenges but ignore the problem of how to incentivize the participation of UAVs. In addition, we optimize the trajectory of UAVs with the information of UEs protected, which is ignored by all of these papers.

Incentive Mechanisms. Incentive mechanism has been widely studied through auction [3], [22], game theory [14], contract theory [23]. Chen et al. [6] propose an iterative matching algorithm to maximize UAV's utility in disaster rescuing. Wang et al. [10] balance between the hovering time and the service capacity in UAV-assist offloading and design a dynamic pricing strategy for UAV profit-maximizing. Ng et al. [3] use UAV swarms to perform federated learning (FL) tasks, where UAVs aim to maximize their individual profit. Wang et al. [24] consider the heterogeneity of UAVs and propose a novel fair and robust federated learning scheme in UAV-aided crowd-sensing for fair incentives and robust model aggregation. Lu et al. [25] focus on package delivery by UAVs and ground vehicles, and propose a multi-model logistics framework to incentivize ground vehicles and plan paths for UAVs. Targeting on crowd-sourcing, Jaimes et al. [22] propose an incentive mechanism for maximizing the coverage of UAVs with a budget constraint. Considering the uncertainty of offloading tasks to UAVs, Apostolopoulos et al. [26] propose a non-cooperative game to maximize each user's satisfaction. In [14], the authors consider a content delivery network cached by UAVs and try to maximize the average revenue of the content provider. Lim et al. [23] consider UAVs and model owners in FL, and propose a contract matching algorithm to maximize the owner's profit. However, the above studies mainly focus on maximizing profit and ignore the efficiency of the system. Furthermore, we schedule the trajectories of UAVs online while protecting the individual location information of UEs.

Primal-dual in MEC. There are some researches using the primal-dual technique to investigate the MEC system. He et al. [27] target on the incentive mechanism design between users and collaborative servers in MEC and use the primal-dual to optimize the social welfare problem with a bounded competitive ratio. Chen et al. [28] focus on pricing the video streaming services in MEC and use the primal-dual to enhance the quality of service. Li et al. [29] try to analyze the social welfare problem in choosing who to execute the task by primal-dual. He et al. [30] propose an auction-based incentive mechanism to optimize the long-term social welfare by primal-dual. None of these studies consider decoupling the time-coupled constraints in the system, and didn't use the primal-dual technique to analyze the impact of time-coupled constraints.

Privacy-preserving in Offloading. To protect the communication channels between UAVs and the BS, Xu et al. [31] consider a dual UAV-assisted MEC system, where the additional UAV acts as a jammer to suppress the eavesdroppers. To protect the information of UAVs, Wei et al. [32] propose a differential privacy-based deep reinforcement learning method in the UAV-assist system. Xu et al. [33] build a distributed ledger based on blockchain to resist the invasion of malicious

UAVs for UAV swarm. To protect the privacy of offloaded data, Liu et al. [34] develop a secure homomorphic encryption framework for processing private data. In [11], the authors propose an efficient asynchronous UAV-assisted FL framework to protect the raw data in training. Wang et al. [35] propose a flight scheduling and task allocation scheme with secrecy rates and non-negative budget constraints to resist eavesdroppers. However, these studies mainly focus on protecting the offloading process and ignore the protection of UEs. Their procedure also introduces the additional privacy cost in the system. We optimize the trajectories of UAVs and protect the privacy of UEs through a new bidding strategy. There is no additional privacy cost in our proposed framework.

III. SYSTEM MODEL

A. System Overview

UAV-assisted Offloading Scenario. As shown in Fig. 1, we consider a UAV-assisted MEC system including a number of user equipments (UEs), a base station (BS), and a mobile network of rotary-wing UAVs. The system involves a time-slotted fashion over T time slots (a.k.a., rounds). Let $\mathcal{X}$ denote the integer set $\{1, 2, \dots, X\}$. Assume there are K UEs and U UAVs. each slot t , there are M_t UEs (where $M_t < K$) requesting to offload tasks for data processing to the BS. All tasks must be processed in one slot due to the delay requirement. In an emergency situation (e.g., sports event, power failure, natural disaster), the BS may be overloaded by too many requests, and needs to hire UAVs to assist offloading. To incentivize the UAVs, a procurement auction is applied where the BS acts as the auctioneer and each UAV submits a series of bids for participation. With all bids collected in a round, the BS determines and pays the selected winners, and schedules UAVs to process tasks.

Offloading Requirement. The offloading tasks from UEs come on the fly. The tasks can be offloaded to UAVs only when these two conditions are satisfied: 1) the UE is connected with a UAV; 2) the connected UAV has sufficient computation capacity to execute the task. If some tasks cannot be handled by UAVs, the BS will upload those tasks to a remote cloud with higher cost. Let $c(\chi)$ be the price function of processing χ units of data in the remote cloud.

UE Privacy-preserving. We assume some UAVs in the swarm are malicious and curious about individual information of UEs.¹ The UE's private information (e.g., the location, offloading requests) infers the UE owner's hobbies or preferences. Once obtained by the malicious UAVs, they can send advertisements to UEs or even threaten the owner's safety. Note that the data of UEs can be protected by other existing methods (e.g., homomorphic encryption), and it is out of the scope in this paper. We aim to protect the location information of UEs, so that it is not obtained by the other UAVs except the scheduled one.

B. Auction Model

Bid Information. We first introduce the bidding language in a traditional scenario without privacy-preserving for UEs. The privacy-preserving auction process is discussed in Sec. IV-C.

¹To get a more specific user profile, the malicious UAVs would honestly conduct the auction process.

At the beginning of each round, UAVs receive the current information of UEs from the BS to compute the feasibility between itself and UEs. Let d_m^t be the data size that UE m wants to offload in slot t , G_m^t be the current position of UE m in slot t (i.e., $G_m^t = (x_m^t, y_m^t)$). F_u^{max} and R_u^{max} are the computation capacity and the maximum coverage radius of UAV u respectively. Each UAV calculates its feasible service sets and submits up to J_u^t bids. Each service set in a bid represents a service scheme. We use $S_u^{t,j}$ to represent the feasible service set that UAV u can cover in its j -th bid at slot t . Each service set is a subset of $\mathcal{M}_t$. UAV u 's j -th bid in round t ($B_u^{t,j}$) is expressed as $B_u^{t,j} = \{b_u^{t,j}, S_u^{t,j}, G_u^{t,j}\}$, where $b_u^{t,j}$ is the claimed cost that UAV u serves the UE set $S_u^{t,j}$ at the location $G_u^{t,j}$.

UAV Properties. The capacity of UAV u 's battery is limited in a single flight, denoted as E_u^{max} . When UAV u serves set $S_u^{t,j}$, there are three types of energy costs generated at slot t : hovering cost, computation cost, and additional propulsion cost,² denoted as $E_u^{h,t}$, $E_u^{s,t}$, $E_u^{f,t}$ respectively. The overall energy cost for UAV u to serve set $S_u^{t,j}$ at slot t is denoted as $E_u^{t,j}$. The hovering cost $E_u^{h,t}$ is a constant for UAV u at every slot. The detailed expressions of computation cost and propulsion cost are $E_u^{s,t} = p_u^s \sum_{m \in S_u^{t,j}} d_m^t$ and $E_u^{f,t} = p_u^f \|G_u^{t,j} - G_u^{t-1}\|$, where p_u^s , p_u^f , and G_u^{t-1} are the unit energy cost of computation, propulsion for UAV u , and the location of UAV u at slot $t-1$ respectively. The moving speed of UAV u is limited and the flying distance of UAV u between two adjacent slots is upper bounded by D_u^{max} .

Decision Variables. After all bids from UAVS received, the BS needs to determine the following variables: i) $x_u^{t,j} \in \{0, 1\}$, which equals 1 if the BS accepts UAV u 's j -th bid $B_u^{t,j}$; ii) $\mathcal{Z}_t \subseteq \mathcal{M}_t$, the set of UEs offloaded to the remote cloud; iii) p_u^t , the payment to UAV u .

Auction Preliminaries. We introduce some definitions in the auction design. Let $v_u^{t,j}$ denote the true cost for UAV u to perform tasks in set $S_u^{t,j}$. The utility of this bid with bidding price $b_u^{t,j}$ is $p_u^t - v_u^{t,j}$ if $x_u^{t,j} = 1$, and 0 otherwise. The BS's utility (u_{BS}) equals $-\sum_{u,j,t} p_u^t x_u^{t,j} - \sum_t c(\sum_{m \in \mathcal{Z}_t} d_m^t)$.

Definition 1: (Truthful Auction). A *truthful* auction guarantees that a different bidding price other than the true value $v_u^{t,j}$ would not increase the utility, i.e., $\forall b_u^{t,j} \neq v_u^{t,j}, u_u^{t,j}(v_u^{t,j}) \geq u_u^{t,j}(b_u^{t,j})$.

Definition 2: (Individual Rationality). An auction is *individual rational* if UAV's utility is non-negative, i.e., $u_u^{t,j}(b_u^{t,j}) \geq 0$.

Definition 3: (Social Welfare, Social Cost). The social welfare equals UAVs' utilities plus the BS's utility. The social welfare is $-\sum_{u,j,t} v_u^{t,j} x_u^{t,j} - \sum_t c(\sum_{m \in \mathcal{Z}_t} d_m^t)$ when payment cancel themselves. Maximizing the social welfare equals to minimizing the social cost $\sum_{u,j,t} v_u^{t,j} x_u^{t,j} + \sum_t c(\sum_{m \in \mathcal{Z}_t} d_m^t)$.

Optimization Objective. In general, UAVs are rational but egocentric. They tend to maximize their own utilities by lying about their true cost. Maximizing social welfare and maximizing the BS's profit are both natural and interesting research problems in real-world auction design. Social welfare maximization prompts the efficiency of the whole eco-system

and everyone has a higher probability to be happy in the long run. Thus, we target the utilities of the entire system (social welfare), which needs to elicit truthful bidding from UAVs. Furthermore, the time-delay and energy consumption are also the popular optimization objectives in UAV related research. Since social cost is a self-defined metric from UAVs, it can also be transferred to the time delay and energy consumption or the combination of these two.

C. Social Cost Minimization Problem

Problem Formalization. Under truthful bidding ($b_u^{t,j} = v_u^{t,j}$), the social cost minimization problem is formulated by the following integer linear problem (ILP):

$$\text{minimize } \sum_{t \in \mathcal{T}} \sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}_u^t} b_u^{t,j} x_u^{t,j} + \sum_{t \in \mathcal{T}} c(\sum_{m \in \mathcal{Z}_t} d_m^t) \quad (1)$$

$$\text{s.t } \sum_{j \in \mathcal{J}_u^t} \sum_{u \in \mathcal{U}} \sum_{m \in S_u^{t,j}} x_u^{t,j} \geq 1, \forall t, \forall m \in \mathcal{M}_t - \mathcal{Z}_t, \quad (1a)$$

$$\sum_{j \in \mathcal{J}_u^t} x_u^{t,j} \leq 1, \forall u, \forall t, \quad (1b)$$

$$\sum_{t \in \mathcal{T}} E_u^{h,t} + \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}_u^t} (E_u^{f,t} x_u^{t,j} + E_u^{s,t} x_u^{t,j}) \leq E_u^{max}, \forall u, \quad (1c)$$

$$x_u^{t,j} \in \{0, 1\}, \mathcal{Z}_t \subseteq \mathcal{M}_t, \forall u, \forall t, \forall j, \forall m \in \mathcal{M}_t. \quad (1d)$$

Constraint (1a) guarantees that all UEs' requests are served. We adopt the XOR-bidding strategy in constraint (1b) which specifies that each UAV wins at most one bid at each round. Constraint (1c) indicates the energy upper bound of each UAV for the whole time span.

Challenges. i) *Intractability.* Even a simplified version of ILP (1) without constraints (1b), (1c) and variable $\mathcal{Z}_t$ is still NP-hard, which is equivalent to the set cover problem [36]. ii) *Online decision-making.* Furthermore, we consider a dynamic offloading scenario. The BS should schedule UAVs on the fly based on the current UE demands and their valuations, which are not known as a priori. Constraint (1c) couples the energy consumption, which is affected at all time slots. iii) *Privacy-preserving.* Protecting the location of UEs is incapable if we permit all UAVs to access information about UEs from the bidding process. An effective mechanism is required to adjust the bidding process with privacy-preserving. Important notations are listed in Table. I for easy reference.

IV. AUCTION DESIGN

A. Overview of Auction Designing

Algorithmic Idea. We design an auction framework, *Ptero*, which solves ILP (1) with privacy-preserved. *Ptero* determines the winning bids and the set of UEs for remote cloud to minimize the social cost with a bounded competitive ratio. The algorithmic idea of *Ptero* is shown in Fig. 2.

- 1) **Problem Decompose.** *Ptero* first decomposes ILP (1) into a set of one-round winner determination problems (WDP) (2) by scaling the energy constraints into the objectives by a carefully designed factor λ_u^t . The detailed implementation of problem decomposition is introduced in Sec. IV-B.
- 2) **Privacy-preserving Pretreatment.** In Sec. IV-C, we show how to protect the individual information of

²Hovering is a special condition of propulsion. Additional propulsion cost is denoted as the extra flying energy cost except hovering cost. We assume that the UAV flies from G_i to G_j without any stop and keeps a constant speed in propulsion for simplicity.

TABLE I
LIST OF NOTATIONS

U	# of UAVs	T	system time-span
K	# of UEs	$\mathcal{X}$	integer set $\{1, 2, \dots, X\}$
J_u^t	# of service sets for UAV u at slot t		
M_t	# of active UEs at slot t		
E_u^{max}	battery capacity of UAV u		
D_u^{max}	max distance between two slots for UAV u		
R_u^{max}	max coverage radius of UAV u		
F_u^{max}	computation capacity of the u		
d_m^t	data size that UE m wants to offload at slot t		
G_m^t	horizontal coordinate of UE m at slot t		
$S_u^{t,j}$	service set in UAV u 's j 'th bid at slot t		
$x_u^{t,j}$	whether or not accept UAV u 's j 'th bid at slot t		
$y_{u,m}^{t,j}$	serve UE m in slot t by UAV u in set j or not		
$\mathcal{Z}_t$	UEs processed in the remote cloud at slot t		
$b_u^{t,j}$	claim cost of UAV u to serve tasks in $S_u^{t,j}$		
$E_u^{h,t}$	hovering energy of UAV u at slot t		
$E_{u,j}^{s,t}$	computation energy of UAV u 's j 'th bid at slot t		
$E_{u,j}^{f,t}$	propulsion energy of UAV u 's j 'th bid at slot t		

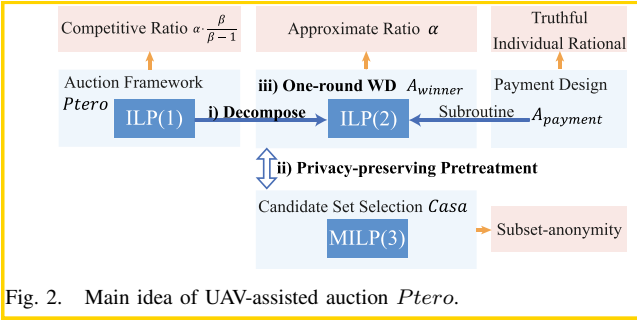


Fig. 2. Main idea of UAV-assisted auction *Ptero*.

UEs by subset-anonymity. We formulate MILP (3) to characterize the matching process between UAVs and UEs. To solve it, *Ptero* modifies the bidding process and provides a privacy-preserving pretreatment algorithm *Casa* based on reverse matching.

- 3) **One-round Winner Determination.** In Sec. IV-D, we show how to determine the winners at each time slot t efficiently. We treat the remote cloud as a special case of UAV for ease of analysis. To solve each WDP (2), *Ptero* calls A_{winner} based on a greedy approach to minimize the total cost at each round with a bounded approximation ratio. The payment is calculated by a subroutine of A_{winner} , $A_{payment}$, to guarantee the truthfulness and individual rationality in the bidding process.

Algorithm Details. We present our auction framework *Ptero* in Alg. IV-B. Line 1 first initializes relative variables for decomposing and winner determination. At time slot t , *Ptero* executes the following steps. In Line 4, *Ptero* calls *Casa* to find the feasible service sets ($S_u^{t,j}$), service locations ($G_u^{t,j}$) for each UAV and the unreachable UAV set $\mathcal{B}_t$. In Line 6, *Ptero* computes the propulsion distance for each service scheme to hide UE's individual information. In Line 7, UAVs prepare bid information. More specifically, once receiving the service sets as well as the propulsion distance from the BS, UAVs claim their costs including bidding price ($b_u^{t,j}$) and additional energy cost ($E_{u,j}^{s,t} + E_{u,j}^{f,t}$). *Ptero* exams the feasibility of each service set in the view of battery capacity and form the grand set $\mathcal{F}^t$, candidate set $\mathcal{C}^t$, and scale their bidding price according to the λ_u^t (which is related to the remaining battery level, and will be discussed in Sec. IV-B) in Lines 9-14. The BS decides the winner set ($\mathcal{A}^t$), the payment ($\mathcal{P}^t$), and cloud

processing set ($\mathcal{Z}_t$) by A_{winner} in Line 15. According to the result returned from A_{winner} , *Ptero* records the best solution which achieves the minimum social cost, and announces the scheduling decision as well as the payment in Line 18. *Ptero* updates the time-energy coupled parameter λ_u^t in Line 17-24. UEs processed in the cloud are announced in Line 26.

Auction Procedure. 1) At the beginning of each round, the BS collects the requests and location of these active UEs. 2) The BS obscures the information of UEs through a sub-algorithm *Casa* and outputs the processed bid metrics (distance and total size of tasks). 3) The BS sends bid metrics to the UAVs. UAVs then tell the BS the bidding price and the additional energy cost of each bid. 4) Using bid information of UAVs, the BS calls a sub-algorithm A_{winner} to determine the winner UAVs and their service sets. 5) The BS calls another sub-algorithm $A_{payment}$ to decide the payment to winners. 6) Selected UAVs start to execute their tasks and receive their remuneration by the end of each round.

B. Problem Decomposition

One-round Problem Formulation. The one-round WDP is defined as follows, which includes the same constraints related to the current time slot from (1), and excludes the time-coupled constraint (1c). We boost the cost $b_u^{t,j}$ into $w_u^{t,j}$ according to the remaining battery capacity of each UAV by a well-designed factor λ_u^t . Thus, the constraint (1c) is considered in the objective function as a scaled cost.

$$\text{minimize } \sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}_u^t} w_u^{(t),j} x_u^{(t),j} + c \left(\sum_{m \in \mathcal{Z}_{(t)}} d_m^t \right) \quad (2)$$

$$\text{s.t. } \sum_{j \in \mathcal{J}_u^{(t)}} \sum_{u \in \mathcal{U}} \sum_{m \in S_u^{(t),j}} x_u^{(t),j} \geq 1, \forall m \in \mathcal{M}_{(t)} - \mathcal{Z}_{(t)}, \quad (2a)$$

$$\sum_{j \in \mathcal{J}_u^t} x_u^{(t),j} \leq 1, \forall u, \quad (2b)$$

$$x_u^{(t),j} \in \{0, 1\}, \mathcal{Z}_{(t)} \subseteq \mathcal{M}_{(t)}, \forall u, \forall j, \forall m \in \mathcal{M}_{(t)}. \quad (2c)$$

The battery capacity limits the service sets that a UAV can cover over T rounds, leading to a different overall social cost when the battery is spent at different rounds. The over-consumption of energy at the early stage may narrow future scheduling decision space. The BS may be consequently forced to choose service sets from higher biddings, which further increases the social cost in the whole time span. The intuition in designing the online auction framework *Ptero* to achieve lower social cost is to reserve a certain amount of battery capacity for future demands. Thus, the scaled cost $w_u^{(t),j} = b_u^{(t),j} + \lambda_u^{t-1}(E_{u,j}^{s,t} + E_{u,j}^{f,t})$ increases with the decrease of a UAV's remaining battery capacity, reducing its chance to win. The adjustment in Line 19 of *Ptero* is carefully computed, such that the energy constraint (1c) is guaranteed over the T rounds of online auctions with a bounded ratio (See Theorem 1 and Theorem 7 for details).

C. Auction Process Modification

To protect UE's information, the auction process is modified.

Definition 4: (Subset-anonymity). Assuming that a set of UEs forms k subsets. The subset-anonymity ensures that an adversary can not identify whether a UE is in one exact subset

Algorithm 1 UAV-assisted offloading auction framework *Ptero***Input:** $G_m^t, d_m^t, G_u^{t-1}, \forall u, m, t$ **Output:** $\mathcal{A}^t, \mathcal{P}^t, \mathcal{Z}_t \forall t$

```

1: Initialize  $\lambda_u^t = 0, \eta_u = 0, \mathcal{C}^t = \mathcal{F}^t = \mathcal{Z}^t = \phi, \forall t, u$ 
2: for  $1 \leq t \leq T$  do
3:   for  $u \in \mathcal{U}, m \in \mathcal{M}_t$  do
4:      $\{S_u^{t,j}, G_u^{t,j}, \mathcal{B}_t\} = Casa\{G_m^t, d_m^t, G_u^{t-1}\}$ 
5:     for  $j \in \mathcal{J}_u^t$  do
6:        $D_u^{t,j} = \|G_u^{t,j} - G_u^{t-1}\|$ 
7:        $\{b_u^{t,j}, E_{u,j}^{s,t} + E_{u,j}^{f,t}\} = bidprocess\{S_u^{t,j}, D_u^{t,j}\}$ 
8:     end for
9:      $\mathcal{F}^t = \mathcal{C}^t = \cup_{u \in \mathcal{U}, j \in \mathcal{J}_u^t} (u, j)$ 
10:    if  $\eta_u + E_{u,j}^{s,t} + E_{u,j}^{f,t} > E_u^{max} - \sum_t E_u^{h,t}$  then
11:       $w_u^{t,j} = b_u^{t,j}; \mathcal{C}^t = \mathcal{C}^t \setminus (u, j)$ 
12:    else
13:       $w_u^{t,j} = b_u^{t,j} + \lambda_u^{t-1}(E_{u,j}^{s,t} + E_{u,j}^{f,t})$ 
14:    end if
15:     $\{\mathcal{A}^t, \mathcal{P}^t, \mathcal{Z}_t\} = A_{winner}\{w_u^{t,j}, S_u^{t,j}, \mathcal{F}^t, \mathcal{C}^t, \mathcal{B}_t\}$ 
16:  end for
17:  for  $(u, j) \in \mathcal{A}^t$  do
18:    Accept UAV  $u$ 's  $j$ 'th bid and schedule  $u$  according to
     $G_u^t, S_u^t$ ; Pay  $p_u^{t,j} \in \mathcal{P}^t$  to the UAV  $u$ 
19:     $\lambda_u^t = \lambda_u^{t-1} \left(1 + \frac{E_{u,j}^{s,t} + E_{u,j}^{f,t}}{\alpha(E_u^{max} - \sum_t E_u^{h,t})}\right) + \frac{b_u^{t,j}(E_{u,j}^{s,t} + E_{u,j}^{f,t})}{\alpha(E_u^{max} - \sum_t E_u^{h,t})^2}$ 
20:     $\eta_u = \eta_u + E_{u,j}^{s,t} + E_{u,j}^{f,t}$ 
21:  end for
22:  for  $u \notin \mathcal{A}^t$  do
23:     $\lambda_u^t = \lambda_u^{t-1}$ 
24:  end for
25:  for  $m \in \mathcal{Z}_t$  do
26:    offload UE  $m$ 's request to the cloud
27:  end for
28:   $\lambda_u = \lambda_u^T, \forall u$ 
29: end for
return  $\mathcal{A}^t, \mathcal{P}^t, \mathcal{Z}_t \forall t$ 

```

of these k subsets. That is, the adversary cannot distinguish an individual UE with a probability higher than $1/k$.

Specifically, we adopt the location subset-anonymity-based strategy to reformulate the bidding service set, which obscures the individual location information of UEs in multiple service areas. The BS calculates the *service set* $S_u^{t,j}$, and the *UAV service location* $G_u^{t,j}$ for UAV u 's serving scheme j . Furthermore, The BS only tells the distance between the service location and the UAV's location ($\|G_u^{t,j} - G_u^{t-1}\|$). With enough service sets, the location of UEs is hidden in the service sets. Thus, the UAV cannot infer the location of a UE. The request tuple provided to UAV u at time t by the BS is expressed as: $\Gamma_u^t = \{\sum_{m \in S_u^{t,j}} d_m^t, \|G_u^{t,j} - G_u^{t-1}\|\}_{j \in \mathcal{J}_u^t}$. The UAV considers its current condition and claims a cost $b_u^{t,j}$ if it wants to accept the set $\Gamma_u^{t,j}$. After jointly considering the information of both UEs and UAVs, the BS decides the winners of each round.

Pretreatment Formulation. Let $\mathcal{B}_t$ denotes the inaccessible UEs caused by distance limitation in slot t (i.e., $\|G_u^{t-1} - G_m^t\| \geq D_u^{max} + R_u^{max}, \forall m \in \mathcal{B}_t, \forall u$), which is a determined in each round. The BS calculates the feasible service sets $S_u^{t,j}$ and the service location $G_u^{t,j}$ for all UAVs in every round. A binary variable $y_{u,m}^{t,j}$ equals 1 if UE m 's task is offloaded to UAV u by scheme j in slot t . Aiming to find all the suitable service sets, the privacy-preserving pretreatment problem in each round is formulated by the following mixed integer linear

Algorithm 2 Candidate Sets Selection *Casa***Input:** $G_m^{(t)}, d_m^{(t)}, G_u^{t-1}, \forall u, m$ **Output:** $S_u^{(t),j}, G_u^{(t),j}, \forall u, j$

```

1: Initialize  $r_u, \mathcal{M}_u = \phi, \mathcal{B}_{(t)} = \phi, \forall u$ 
2: for  $1 \leq u \leq \mathcal{U}$  do
3:   for  $1 \leq m \leq \mathcal{M}_{(t)}$  do
4:     if  $\|G_u^{(t-1)} - G_m^{(t)}\| \leq D_u^{max} + R_u^{max}$  then
5:        $m \rightarrow \mathcal{M}_u$ 
6:        $r_m^{u,circle} = circle(G_m^{(t)}, R_u^{max})$ 
7:     end if
8:   end for
9:    $c_{ij} = circlecross(r_i^u, r_j^u) \forall i, j \in \mathcal{M}_u$ 
10:  label served UEs in  $c_{ij}$ 
11:  while  $\sum_{m \in c} d_m^{(t)} \geq F_u^{max}$  do
12:    roll back the served UEs in this area to its neighbors'
13:  end while
14:  merge  $c_{ij}$  with the same served UEs as  $\hat{c}$ 
15:  for qualified  $\hat{c}$  do
16:     $G_u^{(t),j} = mindist(u, \hat{c})$ 
17:    if  $\|G_u^{(t),j} - G_u^{(t-1)}\| \leq \|D_u^{max}\|$  then
18:       $S_u^{(t),j} = \{m \in \hat{c}\}$ 
19:    end if
20:  end for
21: end for
22:  $\mathcal{B}_{(t)} = \mathcal{M}_{(t)} \setminus \{m \in S_u^{(t),j}, \forall u, j\}$ 
return  $S_u^{(t),j}, G_u^{(t),j}, \mathcal{B}_{(t)}, \forall u, j$ 

```

problem (MILP):

$$\text{maximize } \{S_u^{(t),j}\} \quad (3)$$

$$\text{s.t. } \sum_{j \in \mathcal{J}_u^t} \sum_{m \in \mathcal{U}} y_{u,m}^{(t),j} \geq 1, \quad \forall m \in \mathcal{M}_t - \mathcal{B}_t, \quad (3a)$$

$$\sum_{m \in S_u^{(t),j}} y_{u,m}^{(t),j} d_m^{(t)} \leq F_u^{max}, \quad \forall u, \forall j, \quad (3b)$$

$$\|G_u^{(t),j} - G_u^{(t-1)}\| \leq \|D_u^{max}\|, \quad \forall u, \forall j, \quad (3c)$$

$$y_{u,m}^{(t),j} \|G_u^{(t),j} - G_m^{(t)}\| \leq \|R_u^{max}\|, \quad \forall u, \forall j, \quad \forall m, \quad (3d)$$

$$y_{u,m}^{(t),j} \in \{0, 1\}, G_u^{(t),j} \in \mathbb{R}, S_u^{(t),j} = \{m | y_{u,m}^{(t),j} = 1\}, \quad \forall u, j, m. \quad (3e)$$

Constraint (3a) indicates that all the accessible UEs should be selected at least once. Constraint (3b) confines the offloading capacity of each service set. The flying distance between two slots for UAV u and service coverage between UAVs and UEs are specified in constraints (3c) and (3d) respectively.

Algorithm Idea. To get the feasible solutions of MILP (3), we design an efficient algorithm *Casa* which selects active UEs based on reverse matching. For each UAV, *Casa* first finds out *accessible* UEs for the UAV and calculates the service area that covers these UEs as the possible service set. Then, *Casa* examines the total offloading data size for each set. If the offloading data size of one set exceeds the UAV's computation capacity, *Casa* merges this set with its neighbor sets when the capacity constraint is satisfied. Finally, *Casa* outputs the candidate sets and points their service location where the flying distance is the minimum.

Algorithm Details. The candidate sets selection algorithm *Casa* is presented in Alg. IV-C. Here, $\mathcal{M}_u$ is the *accessible* UEs for UAV u . r_m^u is the valid service area if a UAV u wants to serve UE $m \in \mathcal{M}_u$. c_{ij} is the combined area that covers UE i, j or both. For each UAV u , Lines 4-7 calculate

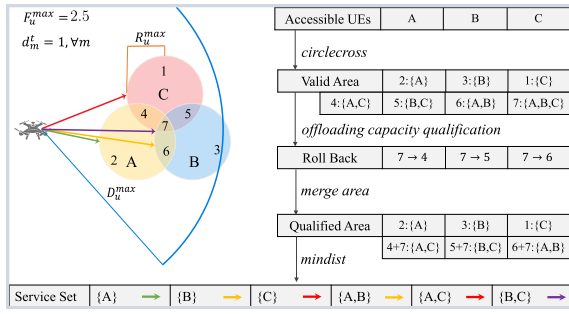


Fig. 3. An example of Casa.

the valid service area r_m^u for accessible UEs. Lines 9-10 find the combined area between UEs and label the covered UEs. Lines 11-13 check the offloading data size for each UE set in the combined area and record the UE sets in the combined area as the same as that of their qualified neighbors. Since an area may roll back to multiple qualified neighbors, line 14 merges these areas with the same UEs for de-duplication. Lines 15-21 determine the service set and the service location for each qualified area $\hat{c}$. Line 17 checks the feasibility of the flying distance between the service location and the current location for the UAV. The inaccessible UE set $\mathcal{B}_{(t)}$ is calculated in line 22.

Example. We illustrate the process of Casa by a simple example shown in Fig. 3. Suppose there are 3 UEs within UAV's coverage, each submits a request (d_m^t) of 1 unit of data. The offloading capacity of UAV u is 2.5 units. Casa first circles the valid service area for each UE, i.e., these three colored circles. Casa then calculates the combined area with different combinations of served UEs, i.e., the 7 numbered areas. Casa next finds out that the requests in area 7 exceed the capacity of UAV u . Area 7 is rolled back to its neighbors whose service set is just one less than its, i.e., 4, 5, 6. With three feasible service sets choices in 7, Casa merges area 7 into its neighbors. After calculating the minimum distance between these areas, the final service set is determined as listed in the figure.

D. Winner Determination Algorithm

Problem Reformulation. We mark the remote cloud as a special case of UAV with the price of introducing an exponential number of service schemes. We mark the remote cloud as the UAV u_0 , which has infinite energy and its service schemes provided in the bidding process are the combination of all the UEs in the current round, i.e., $|\mathcal{J}_{u_0}^{(t)}| = 2^{|\mathcal{M}_{(t)}|}$. The additional energy consumption for offloading a request is 0 so the bidding price is not scaled. The price of service set $S_{u_0}^{(t),j}$ is $w_{u_0}^{(t),j} = b_{u_0}^{(t),j} = c(\sum_{m \in S_{u_0}^{(t),j}} d_m^t)$. We reformulate the problem (2) as the following ILP:

$$\text{minimize } \sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}_u^{(t)}} w_u^{(t),j} x_u^{(t),j} \quad (4)$$

$$\text{s.t. } \sum_{j \in \mathcal{J}_u^{(t)}} \sum_{m \in \mathcal{M}_{(t)}} x_u^{(t),j} \geq 1, \forall u, \forall m \in \mathcal{M}_{(t)}, \quad (4a)$$

$$\sum_{j \in \mathcal{J}_u^{(t)}} x_u^{(t),j} \leq 1, \forall u, \quad (4b)$$

$$x_u^{(t),j} \in \{0, 1\}, \forall u, \forall j. \quad (4c)$$

Constraints (4a) and (4b) are similar to constraints (2a) and (2b) respectively. It is clear that a feasible solution to ILP (4) can be transferred into the feasible solution to the WDP (2).

Dual Problem. Dual technology is a powerful tool to analyze the gap between the optimal and the online solution. To analyze the performance of A_{winner} , we formulate the dual problem by relaxing the integrality constraint (4c) into $0 \leq x_u^{(t),j} \leq 1$, and introduce dual variables $h_m^{(t)}, g_u^{(t)}, k_u^{(t),j}$ to constraints (4a), (4b), and $x_u^{(t),j} \leq 1$, respectively. These dual variables have no practical meaning, only for the convenience of algorithm design and analysis. The dual problem is formulated as follows:

$$\text{maximize } \sum_{m \in \mathcal{M}_{(t)}} h_m^{(t)} - \sum_{u \in \mathcal{U}} g_u^{(t)} - \sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}_u^{(t)}} k_u^{(t),j} \quad (5)$$

$$\text{s.t. } \sum_{m \in S_u^{(t),j}} h_m^{(t)} - g_u^{(t)} - k_u^{(t),j} \leq w_u^{(t),j}, \forall u, \forall j, \quad (5a)$$

$$h_m^{(t)}, g_u^{(t)}, k_u^{(t),j} \geq 0, \forall u, \forall j, \forall m \in \mathcal{M}_{(t)}. \quad (5b)$$

Main Idea. The winner determination algorithm A_{winner} selects schedules iteratively based on a greedy strategy to solve the problem (4). we form $\mathcal{Z}_{(t)}$ with the selection of other UAVs, and decide $x_{u_0}^{(t),j}$ after the UAV selection. We claim a UE is *available* if all these selected UAV service sets do not cover this UE. At each round, A_{winner} selects a UAV with a schedule j which can cover *available* UEs with the lowest *average cost*. If the cost is lower than processing these available UEs in the remote cloud, A_{winner} adds the selected UAV with its corresponding schedule to the winner set. Otherwise, these UEs are chosen for the cloud. This process terminates until all active UEs are served by the winners or the cloud.

Average Cost Let $\mathcal{A}^{(t)} = \{(u_1, j_1), (u_2, j_2), \dots\}$ be a subset of bids submitted at round t , where (u_1, j_1) is UAV u_1 's j_1 'th bidding scheme, $\gamma_m^{A^{(t)}} = \sum_{(u,j) \in \mathcal{A}^{(t)}: m \in S_u^{(t),j}} 1$ denotes the total number of candidate UAVs to serve UE m among all bids in $\mathcal{A}^{(t)}$. The utility of set $\mathcal{A}^{(t)}$ is the number of UEs that has been served in $\mathcal{A}^{(t)}$, defined as $U(\mathcal{A}^{(t)}) = \sum_{m \in \mathcal{M}_{(t)}} \min(\gamma_m^{A^{(t)}}, 1)$. The increased number of served UEs of adding UAV u 's j 'th bid to $\mathcal{A}^{(t)}$ is:

$$\begin{aligned} U_{u,j}(\mathcal{A}^{(t)}) &= U(\mathcal{A}^{(t)} \cup (u, j)) - U(\mathcal{A}^{(t)}) \\ &= \sum_{m \in \mathcal{M}_{(t)}} (\min(\gamma_m^{A^{(t)} \cup (u,j)}, 1) - \min(\gamma_m^{A^{(t)}}, 1)). \end{aligned} \quad (6)$$

The *average cost* of schedule j is $\varphi(m, S_u^{(t),j}) = \frac{w_u^{(t),j}}{U_{u,j}(\mathcal{A}^{(t)})}$. Let $\mathcal{F}^{(t)}$ be the grand set of all bids in round t , and $\mathcal{C}^{(t)}$ be the candidate set of all *valid* bids, such that satisfy XOR-bidding (1b) and UAV battery capacity (1c). At the beginning of A_{winner} , $\mathcal{A}^{(t)}$ is an empty set. Let $\mathcal{L}_{u,j}^{(t)}$ be the current *available* UEs in UAV u 's j 'th bid while selecting bid winners. Then $U_{i,j}(\mathcal{A}^{(t)})$ denotes the number of *available* UEs in that bid.

Algorithm Details. The winner determination algorithm is presented in Alg. IV-D. Line 1 initializes sets and variables. The *while* loop in Line 2-16 iteratively selects UAV bids until all UEs are covered. Specifically, lines 3-4 determine

Algorithm 3 Winner Determination Algorithm A_{winner}

Input: $\{w_u^{(t),j}, S_u^{(t),j}\}, \mathcal{C}^{(t)}, \mathcal{F}^{(t)}, \mathcal{B}_{(t)}, \forall u, m$

Output: $\mathcal{A}^{(t)}, \mathcal{P}^{(t)}$,

- 1: Initialize $\mathcal{A}^{(t)} = \mathcal{P}^{(t)} = \mathcal{L}^{(t)} = \phi, \mathcal{Z}_{(t)} = \mathcal{B}_{(t)}, x_{u,j}^{(t),j} = 0, h_m^{(t)} = 0, g_u^{(t)} = 0, k_u^{(t),j} = 0, \forall m, j, u$ in real
- 2: **while** $U(\mathcal{A}^{(t)}) + |\mathcal{Z}_{(t)}| < |\mathcal{M}_{(t)}|$ **do**
- 3: $(u^*, j^*) = \operatorname{argmin}_{(u,j) \in \mathcal{C}^{(t)}} \frac{w_u^{(t),j}}{U_{u,j}(\mathcal{A}^{(t)})}$;
- 4: $\varphi(m, S_{u^*}^{(t),j^*}) = \frac{w_{u^*}^{(t),j^*}}{U_{u^*,j^*}(\mathcal{A}^{(t)})}, \forall m \in \mathcal{L}_{u^*,j^*}^{(t)}$
- 5: **if** $\sum_{m \in \mathcal{L}_{u^*,j^*}^{(t)}} \varphi(m, S_{u^*}^{(t),j^*}) \leq \Delta c(\sum_{m \in \mathcal{L}_{u^*,j^*}^{(t)}} d_m^t)$ **then**
- 6: $x_{u^*}^{(t),j^*} = 1$
- 7: $p_{u^*}^{(t)} = A_{payment}(C^{(t)}, (u^*, j^*), U_{u,j}(\mathcal{A}^{(t)}))$
- 8: **else**
- 9: $\mathcal{Z}_{(t)} = \mathcal{Z}_{(t)} \cup \{m \in \mathcal{L}_{u^*,j^*}^{(t)}\}$
- 10: **end if**
- 11: $(u^+, j^+) = \operatorname{argmin}_{(u,j) \in \mathcal{F}^{(t)}} \frac{w_u^{(t),j}}{U_{u,j}(\mathcal{A}^{(t)})}$
- 12: $\varphi(m, S_{u^+}^{(t),j^+}) = \frac{w_{u^+}^{(t),j^+}}{U_{u^+,j^+}(\mathcal{F}^{(t)})}, \forall m \in \mathcal{L}_{u^+,j^+}^{(t)}$
- 13: $\mathcal{C}^{(t)} = \mathcal{C}^{(t)} \setminus (\cup_{j \in \mathcal{J}_u^t} (u^*, j))$
- 14: $\mathcal{F}^{(t)} = \mathcal{F}^{(t)} \setminus (u^*, j^*)$
- 15: $\mathcal{P}^{(t)} = \mathcal{P}^{(t)} \cup p_{u^*}^{(t)}; \mathcal{A}^{(t)} = \mathcal{A}^{(t)} \cup (u^*, j^*)$
- 16: **end while**
- 17: $x_{u_0}^{(t),j} = 1$, where $j^\#$ exactly serves $m \in \mathcal{Z}_{(t)}$
- 18: $\max(m), \min(m) = \max, \min_{S_u^{(t),j}} \{ \{ \varphi(m, S_u^{(t),j}) \} \cup \{ \varphi(m, S_u^{(t),j})' \} \}, \forall m, \forall u$ in real
- 19: $\epsilon_m = \frac{\max(m)}{\min(m)}; \epsilon = \max_{m \in \mathcal{M}} \epsilon_m$
- 20: $n_\varphi(m) = \max_{S_u^{(t),j}} \{ \varphi(m, S_u^{(t),j}) \} \forall u$ in real;
 $h_m^{(t)} = n_\varphi(m) / (H_m \epsilon)$
- 21: **for** all $x_{u,j}^{(t),j} = 1$ **do**
- 22: $k_u^{(t),j} = \sum_{m \in \mathcal{L}_{u,j}^{(t)}} (n_\varphi(m) - \varphi(m, S_u^{(t),j})) / (H_m \epsilon)$
- 23: **end for**
- return** $\mathcal{A}^{(t)}, \mathcal{P}^{(t)}, \mathcal{Z}_{(t)}$

the possible winning bid with the lowest average cost at the current iteration and record the corresponding average cost $\varphi(m, S_{u^*}^{(t),j^*})$. Line 5 compares the price for covering the newly added UEs in $\mathcal{L}_{u^*,j^*}^{(t)}$ with the additional cost of adding them to the cloud set. If the UAV is cheaper, the scheme j^* is the winning bid, $x_{u^*}^{(t),j^*}$ is updated to 1. Line 7 calls $A_{payment}$ to calculate the payment of these selected schedules while ensuring truthful bidding. Line 9 updates the remote cloud service set $\mathcal{Z}_{(t)}$ if the cloud is cheaper. Line 11-12 compute the winner average cost $\varphi(m, S_{u^+}^{(t),j^+})'$ if we relax the XOR-bidding constraint and energy restriction to decide the value of dual variables. Line 13 removes all the biddings from winner u^* in the candidate set to satisfy the XOR-bidding rule. Line 14-15 removes the winning bid from grand set $\mathcal{F}^{(t)}$, and adds it to the winner set $\mathcal{A}^{(t)}$. Its relative payment is also added into $\mathcal{P}^{(t)}$. After $\mathcal{Z}_t$ is decided by the *while* iterations, the winner scheme for cloud u_0 is decided in line 17. Lines 18-20 update the dual variables based on the payment to bound approximate ratio, where $H_m = \sum_{m=1}^M \frac{1}{m}$ is the M th harmonic number. The *for* loop in line 21-23 update the dual variable $k_u^{(t),j}$ if the primal constraint is satisfied. (i.e., $x_{u,j}^{(t),j} = 1$).

Payment Design. It is vital to ensure truthful bidding for minimizing social welfare. The basic idea of $A_{payment}$ is to calculate the payment based on the critical bid, i.e., the schedule that has the second smallest bidding value.

$A_{payment}$ is presented in Alg. IV-D. Line 1 finds out the critical bid (u^-, j^-) . Line 2 computes the payment for the selected schedule (u^*, j^*) by the critical value rule [37].

Algorithm 4 Payment Algorithm $A_{payment}$

Input: $C^{(t)}, (u^*, j^*), U_{u,j}(\mathcal{A}^{(t)})$

Output: $p_{u^*}^{(t)}$

- 1: $(u^-, j^-) = \operatorname{argmin}_{(u,j) \in \mathcal{C}^{(t)}: (u,j) \neq (u^*, j^*)} \frac{w_u^{(t),j}}{U_{u,j}(\mathcal{A}^{(t)})}$
- 2: $p_{u^*}^{(t)} = U_{u^*,j^*}(\mathcal{A}^{(t)}) \cdot \frac{w_{u^-}^{(t),j^-}}{U_{u^-,j^-}(\mathcal{A}^{(t)})}$
- return** $p_{u^*}^{(t)}$

V. THEORETICAL ANALYSIS

In this section, we analyze the properties of *Ptero* with its sub-algorithms *Casa*, A_{winner} , and $A_{payment}$. We analyze the correctness, time complexity, truthfulness and individual rationality of the auction, the privacy preserving of *Casa*, the approximate ratio of A_{winner} and the competitive ratio of *Ptero*.

Definition 5: (Approximation Ratio): The approximation ratio of A_{winner} is the upper bound ratio of the objective value of ILP (2) returned by A_{winner} to the optimal objective value of (2) achieved by an optimal algorithm.

Definition 6: (Competitive Ratio): The competitive ratio is the upper-bound ratio of the social cost achieved by *Ptero* to the optimal objective value of the offline WDP in MILP (1).

To better analyze the performance of *Ptero*, we reformulate the original problem (1) without loss as we did in Sec. IV-D. The remote cloud is a special UAV u_0 , whose service set $\mathcal{Z}_t$ is translated to $S_{u_0}^{(t),j}$ in each round. And its additional energy consumption is always 0. Thus, we have $w_{u_0}^{(t),j} = b_{u_0}^{(t),j}$ for u_0 . The reformulated ILP is as follows:

$$\text{minimize } \sum_{t \in \mathcal{T}} \sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}} b_u^{t,j} x_u^{t,j} \quad (7)$$

$$\text{s.t } \sum_{j \in \mathcal{J}} \sum_{u \in \mathcal{U}} \sum_{m \in S_u^{t,j}} x_u^{t,j} \geq 1, \forall t, \forall m \in \mathcal{M}_t, \quad (7a)$$

$$\sum_{j \in \mathcal{J}} x_u^{t,j} \leq 1, \forall u, \forall t \quad (7b)$$

$$\sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}} (E_{u,j}^{s,t} + E_{u,j}^{f,t}) x_u^{t,j} \leq E_u^{max} - \sum_{t \in \mathcal{T}} E_u^{h,t}, \forall u, \quad (7c)$$

$$x_u^{t,j} \in \{0, 1\} \forall u, \forall t, \forall j. \quad (7d)$$

Similar to Sec. IV-D, we use dual technology to analyze the competitive ratio of *Ptero*. We relax its integrality constraints of $x_u^{t,j}$ to $0 \leq x_u^{t,j} \leq 1$ and introduce the dual variables $h_m^t, g_u^t, \lambda_u, k_u^{t,j}$ to constraints (7a), (7b), (7c), and $x_u^{(t),j} \leq 1$, respectively. Note that Prob. (8) is similar to Prob. (5). Most of these variables are related to Prob. (5). Variables $h_m^t, g_u^t, k_u^{t,j}$ in (8) are the online version of the variables in (5) and λ_u is related to the energy factor λ_u^t . These variables also have no

real meaning, and are for easy analysis. The dual problem is formulated as follows:

$$\begin{aligned} \max \quad & \sum_{m \in \mathcal{M}_t} h_m^t - \sum_{u \in \mathcal{U}} g_u^t - \sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}} k_u^{t,j} - \sum_{u \in \mathcal{U}} (E_u^{max}) \\ & - \sum_{t \in \mathcal{T}} E_u^{h,t}) \lambda_u \end{aligned} \quad (8)$$

$$\text{s.t.} \quad \sum_{m \in S_u^{t,j}} h_m^t - g_u^t - k_u^{t,j} - (E_{u,j}^{s,t} + E_{u,j}^{f,t}) \lambda_u \leq b_u^{t,j}, \forall u, \forall j, \quad (8a)$$

$$h_m^t, g_u^t, k_u^{t,j}, \lambda_u \geq 0, \forall u, \forall j, \forall m \in \mathcal{M}_t. \quad (8b)$$

Due to the space limitation, the proofs for all Lemmas and Theorems of such properties mentioned above can be found in the Appendix.

VI. EXTENSION TO GENERAL SCENARIOS

In this section, we extend our model to satisfy more general scenarios where the UE requests to offload a large amount of data such that a single UAV can not process completely in one slot. We first introduce the system model of this general scenario in Sec. VI-A. In Sec. VI-B, we explain the idea of how to modify *Ptero* and its sub-algorithms. The details of the algorithms are presented in Sec. VI-C and the theoretical analysis of these modified algorithms is in Sec. VI-D.

A. System Model

Problem Formulation. Some tasks may be less delay-sensitive but need to offload a large amount of data, e.g., federated learning jobs in UAV-Aided crowd-sensing [24]. These delay-insensitive tasks can be processed by multiple UAVs in multiple slots.

To model the general problem, we introduce some symbols. In slot t , UE m submits the total offloading data size D_m^t and its offloading deadline τ_m^t , i.e., the task with data size D_m^t must be offloaded before slot $t + \tau_m^t$. With all requests collected by the BS, the BS then calculates the suitable service set $\mathcal{J}_u^t$ for bidding. Let $d_m^{t,j}$ be the amount of data that offloaded from UE m at slot t according to scheme j . Note that the previous symbol d_m^t represents the same as $d_m^{t,j}$ since scheme j offloads all the data in one slot. The social cost minimization problem in the general scenario is formulated by the following MILP:

$$\text{minimize} \quad \sum_{t \in \mathcal{T}} \left(\sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}} b_u^{t,j} x_u^{t,j} + c \left(\sum_{m \in \mathcal{Z}_t} D_m^t \right) \right) \quad (9)$$

$$\text{s.t.} \quad \sum_{j \in \mathcal{J}} \sum_{u \in \mathcal{U}} \sum_{m \in S_u^{t,j}} x_u^{t,j} \geq 1, \forall t, \forall m \in \mathcal{M}_t - \mathcal{Z}_t, \quad (9a)$$

$$\sum_{j \in \mathcal{J}} x_u^{t,j} \leq 1, \forall u, \forall t, \quad (9b)$$

$$\sum_{j \in \mathcal{J}} \sum_{u \in \mathcal{U}} y_{u,m}^{t,j} \geq 1, \forall m \in \mathcal{M}_t - \mathcal{Z}_t, \forall t, \quad (9c)$$

$$\sum_{m \in S_u^{t,j}} y_{u,m}^{t,j} d_m^{t,j} \leq F_u^{max}, \forall u, \forall j, \forall t, \quad (9d)$$

$$\|G_u^{t,j} - G_u^{t-1}\| \leq \|D_u^{max}\|, \forall u, \forall j, \forall t, \quad (9e)$$

$$y_{u,m}^{t,j} \|G_u^{t,j} - G_u^t\| \leq \|R_u^{max}\|, \forall u, \forall j, \forall m, \forall t, \quad (9f)$$

$$\sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}} (E_u^{h,t} + E_{u,j}^{f,t} x_u^{t,j} + E_{u,j}^{s,t} x_u^{t,j}) \leq E_u^{max}, \forall u, \quad (9g)$$

$$\sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}} \sum_{t \in [t, t + \tau_t^m]} x_u^{t,j} d_m^{t,j} = D_m^t, \forall m \in \mathcal{M}_t - \mathcal{Z}_t, \quad (9h)$$

$$\begin{aligned} x_u^{t,j}, y_{u,m}^{t,j} &\in \{0, 1\}, G_u^{t,j} \in \mathbb{R}, \mathcal{Z}_t \subseteq \mathcal{M}_t, \\ S_u^t &= \{m | y_{u,m}^{t,j} = 1\}, \forall u, j, m, t. \end{aligned} \quad (9i)$$

Constraint (9a) guarantees that all requests of UEs are offloaded to UAVs (or the remote cloud). Constraint (9b) is the same as that in (3b), which adopts the XOR-bidding for UAVs. Constraint (9c) ensures that all the UEs served by UAVs must be included in the service schemes of UAVs at least once. Constraint (9d) confines that the selected offloading requests in scheme j of UAV u must not exceed its service capacity. The flying distance between two slots for UAV u and the service coverage between UAVs and UEs are restricted by constraints (9e) and (9f). Constraint (9g) indicates the energy consumption of each UAV for the whole time span is limited by its battery capacity. Constraint (9h) ensures that the total offloading request D_m^t is processed during the desired $[t, t + \tau_t^m]$ slots.

Challenges. Except for the challenges mentioned in Sec. III-C, the challenge further escalates in the extended scenario. 1) *Divided requests.* The requests of one UE can be offloaded to multiple UAVs. The solution is fractional. e.g., UAV A processes $\frac{1}{2}$ of the data and UAV B processes the other $\frac{1}{2}$ of the data for a UE. Thus, if we still adopt our modified auction process in Sec. IV, *Casa* may provide countless suitable service sets, which is infeasible for the general scenario. 2) *Multiple slots.* Upon the arrival of large requests, constraint (9h) couples multiple time slots. The problem cannot be decoupled into a series one-round WDP, thus, *Ptero* can not be applied directly in such scenario.

B. Algorithm Design

To tackle the challenges mentioned above, the auction procedure, as well as the auction framework *Ptero*, should be modified. We first modify the auction metric to satisfy the general scenario and the privacy-preserving requirement. We next introduce the modification in *Ptero*.

Auction Metric. To solve the problem (9), the reverse auction metric should be modified while keeping subset-anonymity for UAVs. According to the modified auction process introduced in Sec. IV, *Casa* provides all the possible combinations of suitable service sets, and the UAVs respond with their bidding price to all these schemes. However, such a procedure is incapable in the general scenario. The number of suitable service sets provided to UAVs by *Casa* is countless due to the divided requests mentioned in the challenges. The UAV instead claims its flight cost ($c_u^{f,t}$) per meter and offloading cost ($c_u^{s,t}$) per data size, which makes the problem setting more applicable. Assume that UAV u wins in round t and is scheduled to $G_u^{t,j}$ and serve UEs in $S_u^{t,j}$. We use $b_u^{t,j}(c_u^{f,t}, c_u^{s,t})$ instead of the original $b_u^{t,j}$, which is the cost function of UAV u 's bid in slot t calculated for scheme j . Its cost is calculated as:

$$b_u^{t,j}(c_u^{f,t}, c_u^{s,t}) = c_u^{f,t} \|G_u^{t,j} - G_u^{t-1}\| + c_u^{s,t} \sum_{m \in S_u^{t,j}} d_m^{t,j}$$

Ptero Modification. In practice, the switching cost from one UAV to another UAV is high [38]. If the divided task is processed by cooperatively working UAVs, UAVs need to communicate, bringing high communication cost [39]. Thus, these large-scaled offloading requests prefer to be processed in a single UAV in multiple slots rather than UAVs in cooperation in a single slot. Such a procedure may last for a long period. If the delay can be tolerated, this large-scale offloading request will be processed by the single UAV. Otherwise, this request will be processed by the remote cloud to meet the delay requirement. In the next round, the selected UAV for processing the large request will keep on serving this UE and will not participate in future rounds of the auction until the large request is completed.

Casa Modification. To output countable service sets from candidate sets, *Casa* is modified to *Casa-M*. *Casa-M* mainly adjusts the checking-roll procedure and capacity check procedure.

- 1) The modified checking-roll-back procedure rolls back the overloaded area to the qualified area by excluding the fraction of overloaded UEs to fit the UAV offloading capacity exactly. For example, the overloaded service set $\{A, B, 2C\}$ under offloading capacity 3.5 units are rolled back to $\{A, B, 1.5C\}$ rather than $\{A, B, C\}$.
- 2) The modified capacity check procedure checks whether the UAV is capable for offloading the request during the specified period, i.e., $\tau_t^m F_u^{max} \geq D_m^t$, and outputs these qualified service schemes. *Casa-M* further outputs a list Q_m^t that records the offloading size of the UEs to be processed in this round.

To roll back the overloaded schemes, *Casa-M* only records the schemes with full capacity or schemes with the most UEs. For example, $\{A, B, 0.5C\}, \{A, 0.5B, C\}, \{0.5A, B, C\}$ and reserved in the example in Sec. IV-C, and $\{0.8A, 0.8B, 0.9C\}$ is not a feasible output for *Casa-M* since it does not process the full request of a single UE. The service schemes further pack the qualified area, which is in $O(M^2)$, thus, *Casa-M* outputs schemes in $O(M^2)$.

One-round Determination Modification. To satisfy the new bidding metric, and output the final service set efficiently, we modify the A_{winner} to $A_{winner-M}$ in the payment design and service set update procedure.

- 1) The payment is adjusted to fit in with the new bidding price function, i.e., $b_u^{t,j}(c_u^{f,t}, c_u^{s,t})$ and multiple-slots offloading tasks. It decides the total payment of a multiple-slots offloading task by the current critical value.
- 2) We update the service set in $A_{winner-M}$ to exclude the served UEs in the remaining service set so that the remaining service set exactly fits in with the remaining unserved UEs.

Generally, $A_{winner-M}$ iteratively chooses the service set with the lowest average cost and excludes these served UEs from other sets, then it updates the cost function of these influenced service sets and starts the next iteration until all active UEs are scheduled to the UAVs or the remote cloud. For example, if service sets $\{A, B, 0.5C\}$ and $\{A, 0.5B, C\}$ are the candidates in $A_{winner-M}$, and the former one is chosen in the current iteration, then, the latter one is updated into $\{0.5C\}$ for the next iteration since A and B have been offloaded.

Algorithm 5 UAV-assisted offloading auction framework *Ptero-M*

Input: $G_m^t, D_m^t, G_u^{t-1}, \forall u, m, t$

Output: $\mathcal{A}^t, \mathcal{P}^t, \mathcal{Z}_t \forall t$

```

1: Initialize  $\eta_u = 0, \mathcal{C}^t = \mathcal{Z}_t = \phi, \forall t, u$ 
2: for  $1 \leq t \leq T$  do
3:   for  $u \in \mathcal{U}, m \in \mathcal{M}_t$  do
4:      $\{S_u^{t,j}, G_u^{t,j}, Q_m^t, \mathcal{B}_t\} = \text{Casa-M} \{G_m^t, D_m^t, G_u^{t-1}\}$ 
5:     Exam battery capacity and exclude these incapable
    UAVs.
6:      $\mathcal{C}^t = \cup_{u \in \mathcal{U}, j \in \mathcal{J}_u^t} (u, j)$ 
7:      $\{\mathcal{A}^t, \mathcal{P}^t, \mathcal{Z}_t\} = A_{winner-M} \{c_u^{f,t}, c_u^{s,t}, S_u^{t,j}, G_u^{t,j}, Q_m^t, \mathcal{C}^t, \mathcal{B}_t, d_m^{(t),j}\}$ 
8:   end for
9:   for  $(u, j) \in \mathcal{A}^t$  do
10:    Accept UAV  $u$ 's  $j$ 'th bid and schedule  $u$  according to
     $G_u^t, S_u^t$ ;
11:    Collect battery consumption of this round; Pay
     $p_u^{t,j} \in \mathcal{P}^t$  to the UAV  $u$  after tasks being processed
12:   end for
13:   for  $m \in \mathcal{Z}_t$  do
14:    offload UE  $m$ 's request to the cloud
15:   end for
16: end for return  $\mathcal{A}^t, \mathcal{P}^t, \mathcal{Z}_t \forall t$ 

```

C. Algorithm Details

Details in *Ptero-M*. Similar to *Ptero*, *Ptero-M* in Alg. VI-C first calculates the possible candidate service set by *Casa-M* in line 4. These schemes that violate the battery limitation are excluded in line 5. Line 6 of *Ptero-M* calls $A_{winner-M}$ to decide the auction winners of each round. These winners are scheduled in Line 10. The battery consumption and payment are confirmed in line 11. These tasks scheduled to the remote cloud are announced in line 14.

Details in *Casa-M*. We modified *Casa* into *Casa-M*. A new variable Q_m^t is introduced to record the data size of UE m that the system should offload in round t . Line 2 excludes these multiple-slots served UEs in the single-slot candidate selection. These UEs are continuously served by the UAV of their first-round chosen. The main modification is on the candidate formation procedure in lines 4~16. Line 6 reschedules these overloaded sets to satisfy constraint (9d). Lines 8~14 decide the multiple-slots offloading request selection schemes. If the service period τ_t^m is sufficient for a single UAV to serve the UE, the UAV is selected to offload as much as it can. If not, *Casa-M* drops this UE for the UAV, if none of these UAVs can serve this UE, this UE is served by the remote cloud. The UEs that are served in a single slot is calculated the same in *Casa*. Specifically, line 7 of *Casa-M* examines these overloaded service sets in polynomial time, a service set could at most trace back to the other $O(M^2)$ service sets. In *Casa-M*, there will not be more than two UE with the fractional offloading requests in a qualified service set. Thus, the procedure of line 7 is not combinatorial. The time complexity of *Casa-M* is $O(UM^3)$.

Details in $A_{winner-M}$. The decision of a suitable combination of serving UAVs is decided in $A_{winner-M}$. $A_{winner-M}$ records the remaining requests of each active UE after a winner is decided in each iteration and changes the schemes related to these UEs in the candidate sets. $A_{winner-M}$ ends the *While*-loop if all the requests of UEs (Q_m^t) in this round are assigned.

Algorithm 6 Candidate Sets Selection Algorithm *Casa-M*

Input: $G_m^{(t)}, D_m^{(t)}, G_u^{t-1}, \forall u, m$

Output: $S_u^{(t),j}, G_u^{(t),j}, Q_m^{(t)}, \mathcal{B}_{(t)}, \forall u, j, m$

```

1: for  $1 \leq u \leq \mathcal{U}$  do
2:   If UAV  $u$  has already assigned to serve the UE  $m$  in a
   period, remove the UE  $m$  and UAV  $u$  from candidates.
3:   Lines 3~9 in Casa
4:   for UE sets in  $c_{ij}$  do
5:     Check offloading capacity in each service area.
6:     if  $\sum_{m \in c_{ij}} d_m^{(t)} \geq F_u^{max}$  then
7:       Take turns to fill the service capacity by UEs in  $c_{ij}$ 
       and form the service schemes as follows
8:       if  $\tau_{(t)}^m F_u^{max} \geq D_m^{(t)}$  then
9:          $Q_m^{(t)} = F_u^{max}$  //offload by single UAV, serve it
       until finish
10:      else
11:        move this UE to the remote cloud
12:      end if
13:    else
14:      Form the service schemes,  $Q_m^{(t)} = D_m^{(t)}, \forall m \in c_{ij}$ 
15:    end if
16:  end for
17:  Lines 14~20 of Casa
18: end for
return  $S_u^{(t),j}, G_u^{(t),j}, \mathcal{B}_{(t)}, Q_m^{(t)} \forall u, j$ 

```

Algorithm 7 Winner Determination Algorithm *A_{winner-M}*

Input: $\{c_u^{f,(t)}, c_u^{s,(t)}, S_u^{(t),j}, G_u^{(t),j}, Q_m^{(t)}, \mathcal{C}^{(t)}, \mathcal{B}_{(t)}, d_m^{(t),j}\}, \forall u, m, j$

Output: $\mathcal{A}^{(t)}, \mathcal{P}^{(t)}, \mathcal{Z}_{(t)}$

```

1: Initialize  $\mathcal{A}^{(t)} = \mathcal{P}^{(t)} = \phi, \mathcal{Z}_{(t)} = \mathcal{B}_{(t)}, x_u^{(t),j} = 0, \forall m, j, u$ 
2: while  $Q_m^{(t)} \neq \phi$  do
3:    $b_u^{(t),j} = c_u^{f,(t)} \|G_u^{(t),j} - G_u^{t-1}\| + c_u^{s,(t)} \sum_{m \in S_u^{(t),j}} d_m^{(t),j}, \forall u, j$ 
4:    $(u^*, j^*) = \argmin \frac{b_u^{(t),j}}{V(S_u^{(t),j^*})}, \forall u, j$ 
5:    $\varphi(m, S_u^{(t),j^*}) = \frac{b_{u^*}^{(t),j^*}}{V(S_{u^*}^{(t),j^*})}, \forall m \in S_{u^*}^{(t),j^*}$ 
6:   if  $\sum_{m \in S_{u^*}^{(t),j^*}} \varphi(m, S_{u^*}^{(t),j^*}) \leq \Delta c(\sum_{m \in S_{u^*}^{(t),j^*}} d_m^{(t),j^*})$ 
   then
7:      $x_{u^*}^{(t),j^*} = 1$ 
8:      $p_{u^*}^{(t)} = A_{payment}(b_{u^*}^{(t),j^*}, S_{u^*}^{(t),j^*}, (u^*, j^*))$ 
9:   else
10:     $\mathcal{Z}_{(t)} = \mathcal{Z}_{(t)} \cup \{m \in S_{u^*}^{(t),j^*}\}$ 
11:  end if
12:   $\mathcal{C}^{(t)} = \mathcal{C}^{(t)} \setminus (\cup_{j \in J_{u^*}^{(t)}} (u^*, j))$ 
13:   $\mathcal{A}^{(t)} = \mathcal{A}^{(t)} \cup (u^*, j^*), \mathcal{P}^{(t)} = \mathcal{P}^{(t)} \cup p_{u^*}^{(t)}$ 
14:   $Q_m^{(t)} = Q_m^{(t)} - d_m^{(t),j^*}, \forall m \in S_{u^*}^{(t),j^*}$ 
15:   $S_u^{(t),j} - > D_m^{(t)} - d_m^{(t),j^*}, \forall m \in S_{u^*}^{(t),j^*}$ 
16: end while
17:  $x_{u_0}^{(t),j^\#} = 1$ , where  $j^\#$  exactly serves  $m \in \mathcal{Z}_{(t)}$ 
return  $\mathcal{A}^{(t)}, \mathcal{P}^{(t)}, \mathcal{Z}_{(t)}$ 

```

To better illustrate the adjustment in $A_{winner-M}$, a new function $V(S_u^{(t),j})$ is introduced, which represents the percentage of data size for these to be served UEs in the service scheme $S_u^{(t),j}$. For example, there are two UEs A and B, each requests 12 units of data for offloading, scheme S processes 10 units of data for A and 6 units of data for B. $V(S) = \frac{10}{12} * 12 + \frac{6}{12} * 12 = 18$. Lines 2~16 of $A_{winner-M}$ iteratively

Algorithm 8 Payment Algorithm *A_{payment-M}*

Input: $b_u^{t,j}, S_u^{(t),j}, (u^*, j^*)$

Output: $p_{u^*}^{(t)}$

```

1:  $(u^-, j^-) = \argmin_{(u,j) \neq (u^*, j^*)} \frac{b_u^{(t),j}}{V(S_u^{(t),j})}$ 
2: if The UE requires multiple slot offloading then
3:    $p_{u^*}^{(t)} = D_m^{(t)} \cdot \frac{b_{u^-}^{(t),j^-}}{V_{u^-,j^-}(\mathcal{A}^{(t)})}$ 
4: else
5:    $p_{u^*}^{(t)} = V(S_{u^*}^{(t),j^*}) \cdot \frac{b_{u^-}^{(t),j^-}}{V_{u^-,j^-}(\mathcal{A}^{(t)})}$ 
6: end if
return  $p_{u^*}^{(t)}$ 

```

select UAV bids until all requests of UEs in this round are offloaded. Line 3 calculates the total bidding price of each scheme and Line 4 selects the most efficient schemes. Line 5 is the average cost of the winner's schedule. If the average cost is smaller than that in the remote cloud, these UEs in the winner schedule are assigned to the winner UAV. Otherwise, the remote cloud will offload them. Lines 12~15 update the decision variables as well as the remaining service sets. The remaining service set is updated to exclude the served part of the UE in the selected UAVs.

Since the auction metric has changed, the payment of UAVs should also be recalculated. We still adopt the critical value in the payment design. The bidding price should be fixed at the beginning even if the UAV wins a multiple-slots offloading request. Otherwise, the UAV may increase its price for higher profit since it has been assigned in future slots. The scale factor of payment for its future movement and offloading is determined in the first slot by the critical value. The detailed expression is shown in $A_{payment-M}$.

The payment algorithm in Alg. VI-C is similar to Alg. IV-D. It first finds out the critical bid (u^-, j^-) in line 1. The payment of multiple slot offloading requirements is calculated based on the current critical value [37] in line 3. Line 5 is the payment for the single-slot schemes.

D. Theoretical Analysis

In this section, we analyze the performance of *Ptero-M* and its sub-algorithms. We prove that *Ptero* achieves feasibility, truthfulness, individual rational and subset-anonymity. Due to the space limitation, the proofs can be found in the appendix.

VII. PERFORMANCE EVALUATION

A. Evaluation Setup

Parameter Settings. We consider a UAV-assisted MEC network with 15 UAVs ($U=15$) and 55 UEs ($K=55$), and the percent of active UEs in each slot (M_t) is set to 80%. We utilize a recently-published bike trace [40], which includes GPS coordinates of a million shared bicycles in Shanghai, China. We select a specific region in 3 km*5 km and assume that a UE is carried by bike. The length of each slot is 2 minutes and the entire system lasts for 45 slots (T). In each slot, the task size ($d_m^{(t)}$) of an active UE is set within [5,10] Mb. The configuration of UAVs is set according to the properties of DJI Mavic 2 Pro [41]. Specifically, the battery capacity (B_u^{max}) is 60 Wh. The maximum flying speed is 20 m/s. Since

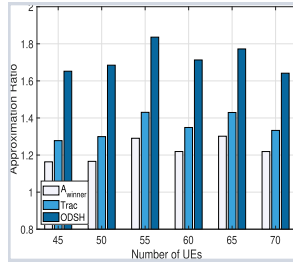


Fig. 4. Approximate ratio of A_{winner} , $Trac$ and $ODSH$ under different numbers of UEs.

the trace updates coordinate every 40 seconds, the maximum travel distance (F_u^{max}) between slots is 800m. The hovering energy cost ($E_u^{h,t}$) is 4 kJ in a slot. The additional unit propulsion energy cost (p_u^f) is 4 J/m. The maximum offloading capacity (D_u^{max}) is 40 Mb/slot, and the transmit power (p_u^s) is 20 dBm. To stabilize the offloading process, we limit the radiation radius (R_u^{max}) to 400 m. Suppose that the initial coordinates of UAVs are randomly distributed in this square. The bidding price of offloading a unit size of the data for UAV is uniformly distributed in $[10,16]$. Since the remote cloud has a large computation capacity, the price would hardly change dynamically, we set the price function of the remote cloud to grow linearly with a unit price of 30.

Benchmark Algorithms: We compare *Ptero* with A_{winner} to four benchmark algorithms.

- **$A_{pricing}$ [10]:** which scales the price according to its overall expectation of profit in choosing to serve a task or stay still. For the one-round choosing strategy, we adopt the same as that in A_{winner} .
- **Greedy:** a greedy algorithm that selects winners using their original bidding price $b_u^{t,j}$ rather than $w_u^{t,j}$. The unqualified bids are also excluded in the current round. Its one-round choosing strategy is the same as A_{winner} .
- **ODSH [16]:** UAVs are scheduled to their nearest service set for energy saving in each round. If UAVs have no power, they are excluded in the future rounds. We compare its one-round performance with A_{winner} and the overall result with *Ptero*.
- **Trac [15]:** a one-round algorithm adopted in location-aware crowdsourcing without the XOR-bidding constraint. It calculates the effectiveness of a service set based on the total covered UEs rather than the number of *available* UEs, and selects the most effective service sets iteratively.

B. Performance of One-Round Auction A_{winner}

Approximation Ratio. Fig. 4 shows the approximation ratio of A_{winner} , $Trac$, and $ODSH$ under different numbers of UEs. All of them perform well with the increase in the number of UEs. However, A_{winner} always outperforms the other two, because A_{winner} is able to select more cost-efficient service schemes to cover the UEs.

Social Cost and Payment. In Fig. 5, we compare the social cost and payment under different bidding price distributions: UNI (a uniform distribution with unit cost in $[10,16]$), NORM (a normal distribution with the mean unit cost and standard deviation of 13 and 3 respectively), and EXP (an exponential distribution with a mean cost of 16). When the number of UEs increases, the BS must hire more UAVs, incurring a higher social cost. The social cost of the exponential distributions is

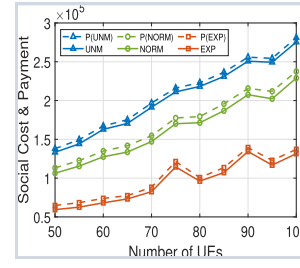


Fig. 5. Social cost & payment with different numbers of UEs and price distributions.

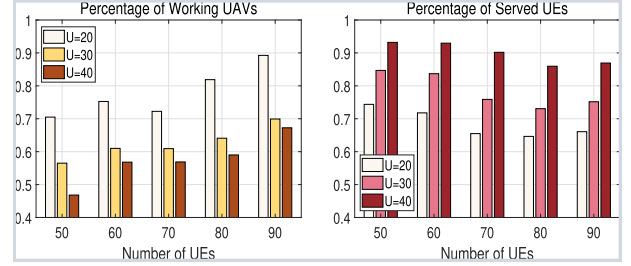


Fig. 6. Percentage of working UAVs and served UEs under different numbers of UAVs and UEs.

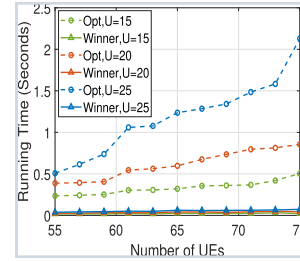


Fig. 7. The average running time of A_{winner} and *Optimal* under different numbers of UEs and UAVs.

lower than the other two because it generates a lower unit price with a larger probability. The payment in these three distributions is always a bit higher than the social cost as $A_{payment}$ computes the payment no smaller than its cost.

System Satisfaction and Time Complexity. In Fig. 6, we examine the percentage of working UAVs and served UEs by UAVs in various conditions. The percentage of working UAVs increases with more UEs and sharply decreases when more UAVs participate. The percentage of served UEs is always larger than 60% in all conditions and changes dynamically. When there are more UEs distributed in the area, the UAV cannot cover many of them due to its limited service area, which leads to a decrease in served UEs and an increase in working UAVs.

Fig. 7 illustrates the complexity of A_{winner} under different input scales. We apply the *tic* and *toc* functions in MATLAB to measure the main body of the program. We repeat 20 times on a laptop (Intel i7-9750H/16GB RAM) and plot the average value. The running time of A_{winner} is only 72ms even with input on a large scale ($U = 25, M = 75$). It can be observed that A_{winner} grows linearly with the growing number of UEs and UAVs. We also plot the running time of the optimal algorithm *intlinprog* in MATLAB, which is significantly larger than ours.

C. Performance of Online Auction *Ptero*

Execution Time. Fig. 8 shows the average execution time of *Ptero* for 45 slots. The execution time is longer than the sum of

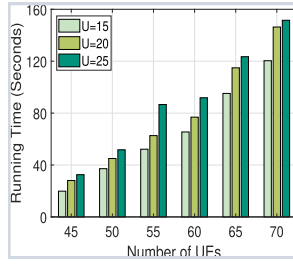


Fig. 8. The average execution time of *Ptero* under different numbers of UEs and UAVs.

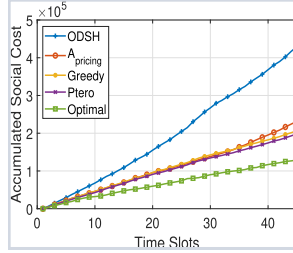


Fig. 9. Social cost of five algorithms accumulated in slots.

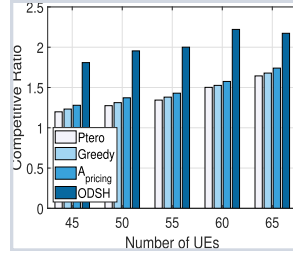


Fig. 10. Competitive ratio of four algorithms under different numbers of UEs.

A_{winner} plotted in Fig. 7, since *Ptero* also includes *Casa* and variable updating process. The time is still acceptable even when we include 70 UEs and 25 UAVs in the system. The running time grows linearly with the increase of the number in UAVs and UEs, which confirms our time complexity analysis.

Social Cost. We analyze the accumulation of social cost in the entire time span of these five algorithms in Fig. 9. All of them grow near-linear as time goes on. *ODSH* performs the worst since it only considers the shortest path to reserve energy. The gap between *Ptero*, *greedy*, and *A_pricing* starts to increase in around slot 30. At that time, some UAVs may have run out of energy under the schedule of *greedy* and *A_pricing*, leading to less number of available UAVs to schedule. Although the final gap between *greedy* and *Ptero* is around 3%, *Ptero* has always outperformed in all slots, proving the efficiency of decomposition.

Competitive Ratio. Fig. 10 illustrates the competitive ratio of *Ptero*, *Greedy*, *A_pricing*, and *ODSH* with different numbers of UEs. All of them perform worse with the increase of UEs. The rank is the same as that in Fig. 9, where *Ptero* performs the best in these four online algorithms. we observed that there is only a small performance loss in *Ptero* compared with A_{winner} in Fig. 4, which confirms the analysis of decomposition loss in Theorem 7.

Performance of UAVs. Fig. 11 shows the remaining power of three groups of UAVs with different configurations after performing all rounds of tasks under different numbers of

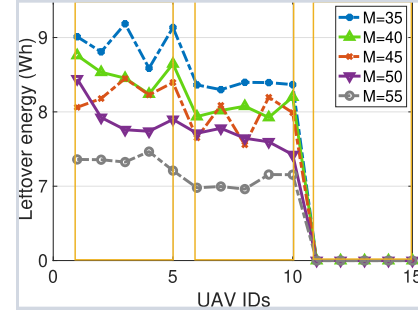


Fig. 11. The leftover energy of 15 UAVs in three capacity groups under different number of UEs.

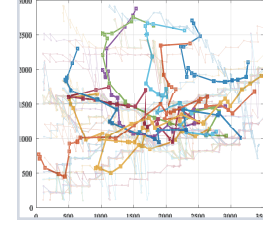


Fig. 12. 10 UAV traces with 55 UEs in 45 slots.

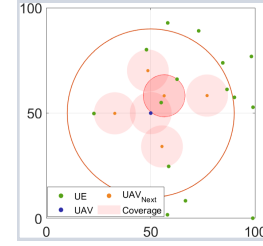


Fig. 13. An example of *Casa* with 1 UAV and 15 UEs.

UEs. The difference between these three groups is the battery capacity, i.e., 66.7Wh, 60Wh (Default), and 50Wh. The last five UAVs with less energy fail to perform tasks. The system tends to allocate more tasks to the UAVs with higher capability so that the leftover energy of these two groups is close. The more UEs to serve the less energy left after rounds finish. We plot the overall traces of UEs and the corresponding trajectory traces of UAVs returned by *Ptero* in Fig. 12. Specifically, the 55 lighter lines indicate the trace information of UEs and the 10 darker lines are these UAVs. The UAVs tend to fly to areas where more UEs are active. Since its coverage is 400m, most of these UEs are served by UAVs. Most of these UAVs travel around the map instead of hovering in a certain area, which avoids the local optimum decision for a single UAV. Fig. 13 shows the result of *Casa* with 1 UAV and 20 UEs distributed in 100*100 squares. The large orange circle is the maximum service area ($D_u^{max} + R_u^{max}$) for the UAV. These small pink circles and the orange dots are the coverage area and service location for a service set. Note that the darker circle in the northeast of the UAV covers more than one UE with minimum trajectory distance. The UAV stays still if a UE has been within its service coverage.

Information Leakage. We assume there is one malicious UAV in the system by default. The percentage of information leakage is defined as the ratio between the location of UEs served by this malicious UAV and all the UE movement information in all slots. We analyze the influence of different service capacity (D_u^{max}) of and maximum moving distance

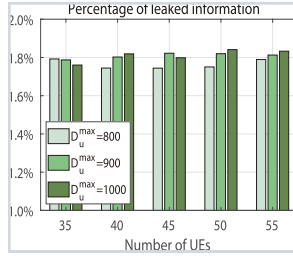


Fig. 14. The percentage of leaked information under different number of service capacity and moving distance.

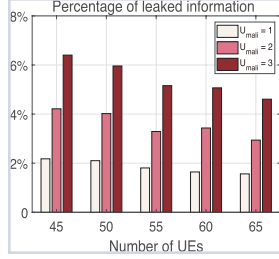


Fig. 15. The percentage of leaked information under different number of UEs and malicious UAVs.

(F_u^{max}) of UAVs and record the percent of information leakage in Fig. 14. It can be observed that the percentage of leaked information slightly grows with the increasing service capacity of the malicious UAV. The more data it offloads, the more probability it connects to more UEs. The maximum moving distance does not have a major influence on the percentage of leaked information. We further test the relationship between the number of malicious UAVs in the system and the percentage of leaked information in Fig. 15. The percentage decreases as more UEs participate. Although the leaked information grows with more malicious UAVs in the system (U_{mali} represents the number of malicious UAVs in the figure), it still stays relatively low (under 7%).

D. Performance of Ptero-M

Extended Settings. The auction metric is modified to the flight cost ($c_u^{f,t}$) per meter and offloading cost ($c_u^{s,t}$) per data size. Similar to [42], we set $c_u^{f,t}$ to $[0.05, 0.1]$ and $c_u^{s,t}$ to $[5, 10]$. The expectation of the number of delay-insensitive UEs is set to 5% of the total active UEs in each slot. The execution time slot (τ_t^m) for UEs is set to $[1, 6]$, and the total data size (D_t^m) is set to $\tau_t^m * d_t^m$, i.e., $\tau_t^m = 1$ is a delay-sensitive UE and $\tau_t^m > 1$ is delay-insensitive UE. The other parameters remain the same as we defined in Sec. VII-A.

In the general scenario, these two offloading algorithms are introduced for comparison.

- Non-Divided-Offloading (NDO): Similar to *Ptero*, the request from a UE is offloaded to a single UAV. The UE with a large amount of data is also offloaded by one UAV through multiple slots, if there is no satisfied UAV, the request is offloaded to the remote cloud.
- ODSH [16]: The BS allocates the UAV to the closest scheme for energy saving. The offloading decision for the delay-insensitive UEs is decided in the first slot and will not be preempted in later rounds.

Social Cost. We examine the social cost under different numbers of UEs and the percentage of delay-insensitive UEs to the total active UEs in Fig. 16. The social cost grows with more delay-insensitive UEs in the system since they occupy

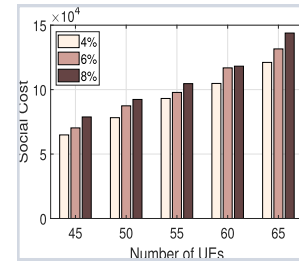


Fig. 16. The social cost achieved by *Ptero-M* under different number of UEs and delay-insensitive UEs.

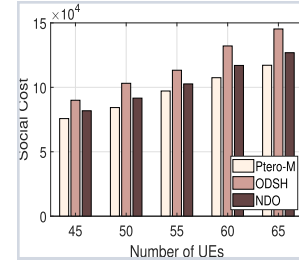


Fig. 17. The social cost achieved by *Ptero-M*, NDO and ODSH under different number of UEs.

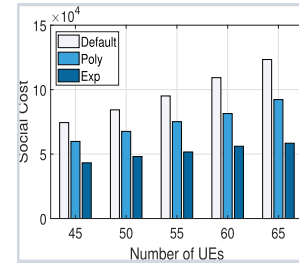


Fig. 18. The social cost with different cost function for the remote cloud.

more UAVs in more slots so that the UAVs can not fly to serve the others. We next examine the social cost achieved by these three algorithms under different numbers of UEs in Fig. 17. *Ptero-M* achieves a better performance than the other two since it combines the offloading data and the flying distance into consideration. Due to the fractional offloading scheme, *Ptero-M* also performs better than NDO.

In Fig. 18, we test the influence of the price function of the remote cloud $c(\chi)$ on the total social cost. The default setting is to grow linearly with the unit price of 30, i.e., $c(\chi) = 30 * \chi$. The polynomial function (abbreviated as poly) is defined as $c(\chi) = \frac{\chi^2}{U}$, and the exponential function (abbreviated as exp) is defined as $c(\chi) = M * U^{x_M}$. All of them grow steadily with the number of UEs increasing, and the exponential function grows slower than the other two since it grows slowly when the size of offloaded data is small.

Performance of UAVs. We analyze the working condition of these UAVs under different numbers of UEs and the percentage of delay-insensitive UEs to the total active UEs in Fig. 19. The UAVs are divided into single-slot (the UAVs that serve the normal UEs), multiple-slots (the UAVs that serve the delay-insensitive UEs), and idle (the UAVs that hover still in the slot). The three bars in each number of UEs represent the percentage of delay-insensitive UEs (4%, 6%, and 8% respectively). The percentage of working UAVs (single-slot+multiple-slots) keeps steady in the same condition. The working rate slightly decreases with more UEs participating.

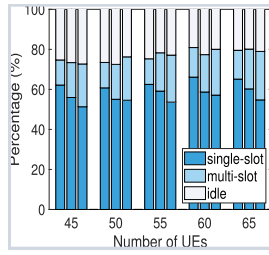


Fig. 19. The UAV working condition in different number of UEs and expected delay-insensitive UEs.

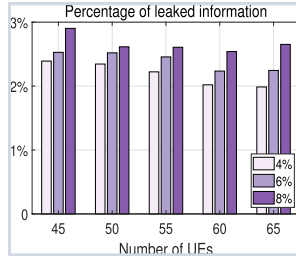


Fig. 20. The percentage of leaked information under different number of UEs and expected delay-insensitive UEs.

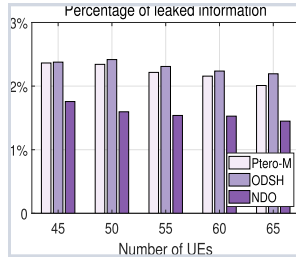


Fig. 21. The percentage of leaked information under different number of UEs and three algorithms.

The multiple-slots UAV rate grows when more UEs are delay-insensitive.

Information Leakage. We record the leaked information in the general scenario. Fig. 20 shows the percentage of leaked information from *Ptero-M* under different numbers of UEs and the percentage of delay-insensitive UEs to the total active UEs. Compared to Fig. 15, the percentage of leaked information slightly grows since we permit fractional offloading and persistent offloading for the delay-insensitive UEs. The rate also increases as the percentage of delay-insensitive UEs grows. The more delay-insensitive UEs, the higher probability the malicious UAV can obtain information from them. We also compare the information leakage from these three algorithms in Fig. 21. Most of the information is protected from *Casa-M*, and these three behave differently in winner selection, which causes different percentages of information leakage. NDO behaves the best among these three since the divided offloading request is forbidden in this strategy, which reduces information leakage. *Ptero-M* and ODSH behave near the same.

VIII. CONCLUSION

UAV complements MEC by providing flexible and low-latency services for UEs in the era of 5G. In this paper, we study online mechanism design for task offloading in UAV-assisted MEC, for scheduling offloading tasks on the fly and incentivizing UAV's participation. A reverse auction

framework, *Ptero*, is proposed, with a goal of social cost minimization. Different from existing literature, *Ptero* determines UAV's trajectory and UE-UAV association for task offloading under battery capacity constraint, while protecting UE's information. Through rigorous theoretical analysis, we prove that *Ptero* is truthful, individual rational, computational efficient, privacy preserving and achieves a good competitive ratio. Trace-driven simulations further show that *Ptero* outperforms four baselines and has near-to-optimal performance.

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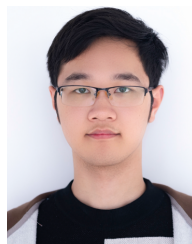
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